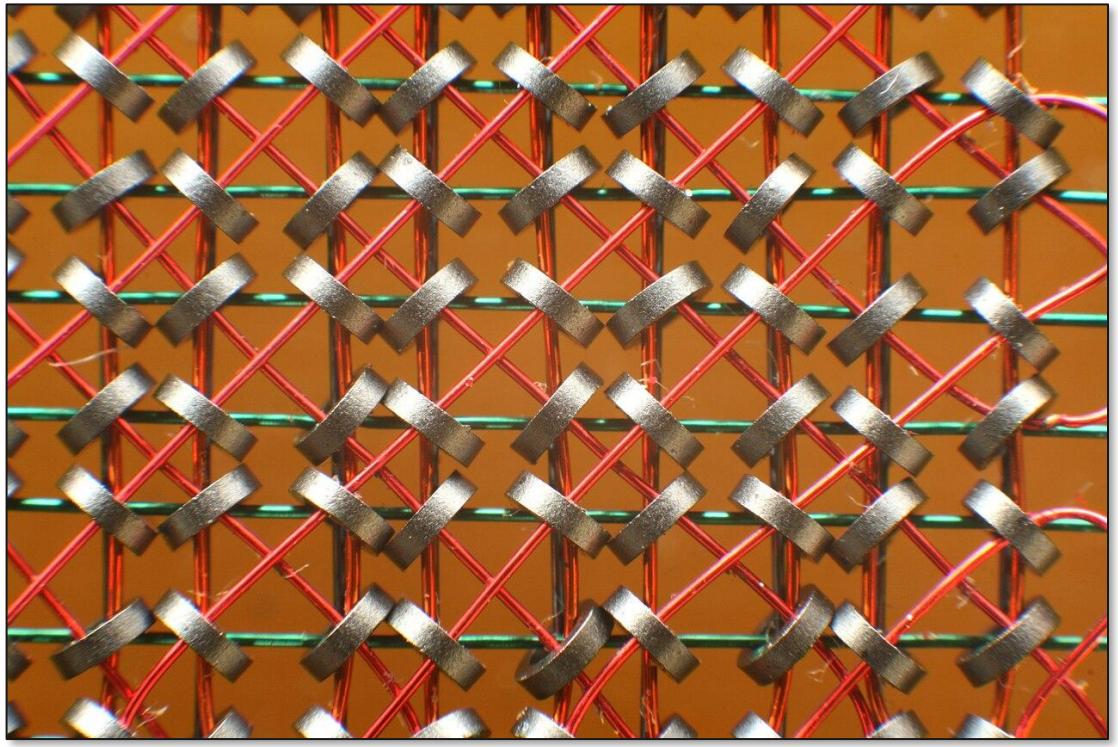
EE426 / 626 Microprocessor Applications

A Brief Trip Down Memory Lane

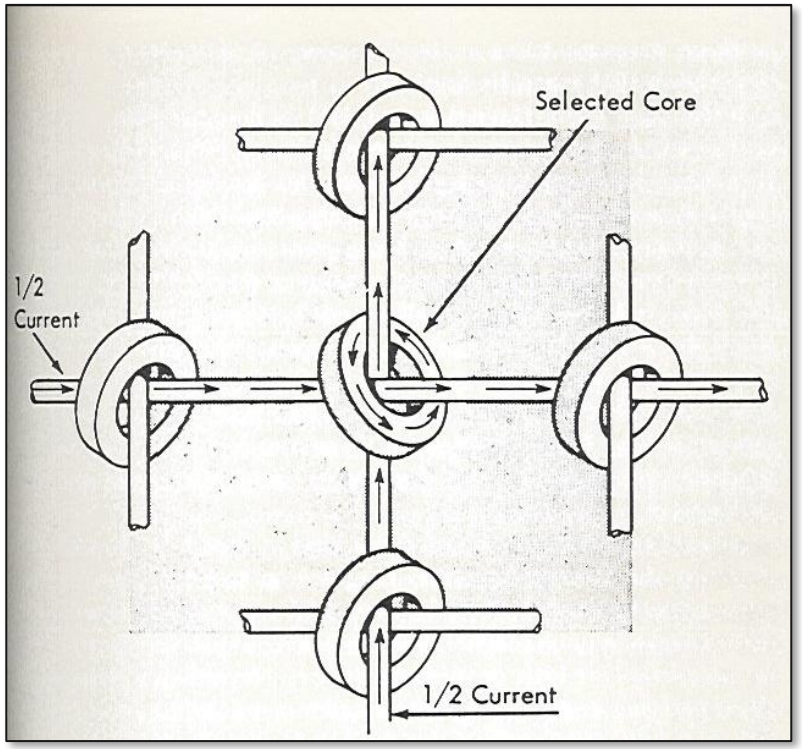


EE426 / 626 Microprocessor Applications

Magnetic Core Memory



By Konstantin Lanzet - received per EMailCamera: Canon EOS 400D, CC BY-SA 3.0, https://commons.wikimedia.org/w/index.php?curid=7025574



Fair use, Columbia University
www.Columbia.edu/cu/computinghistory/core.html

EE426 / 626 Microprocessor Applications

IBM 2361 Large Capacity Storage (LCS) Module

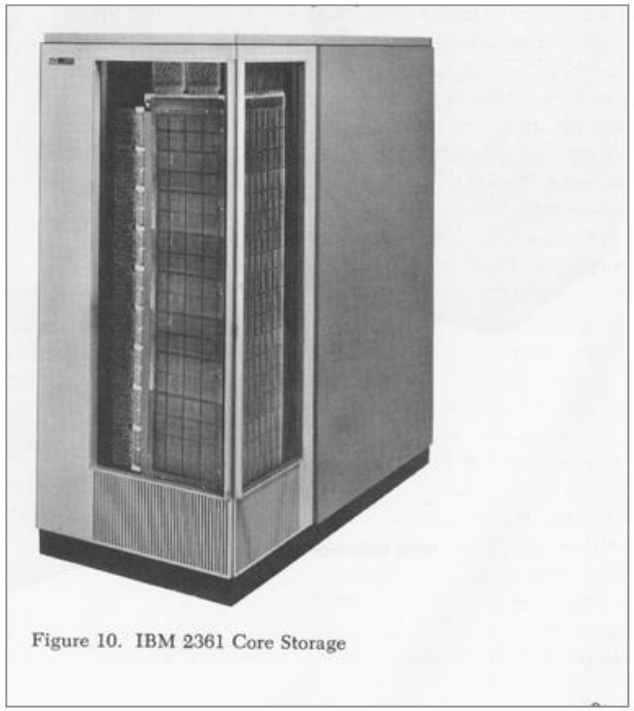
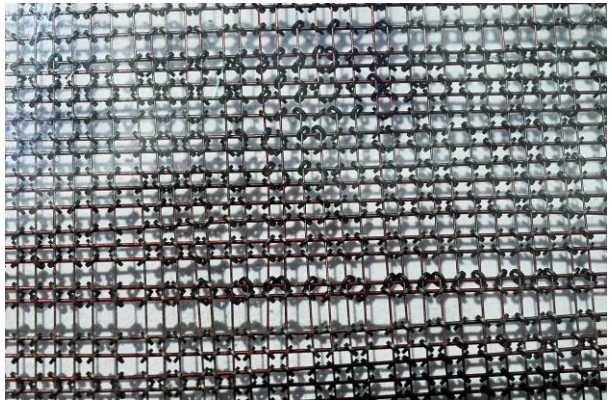


Figure 10. IBM 2361 Core Storage

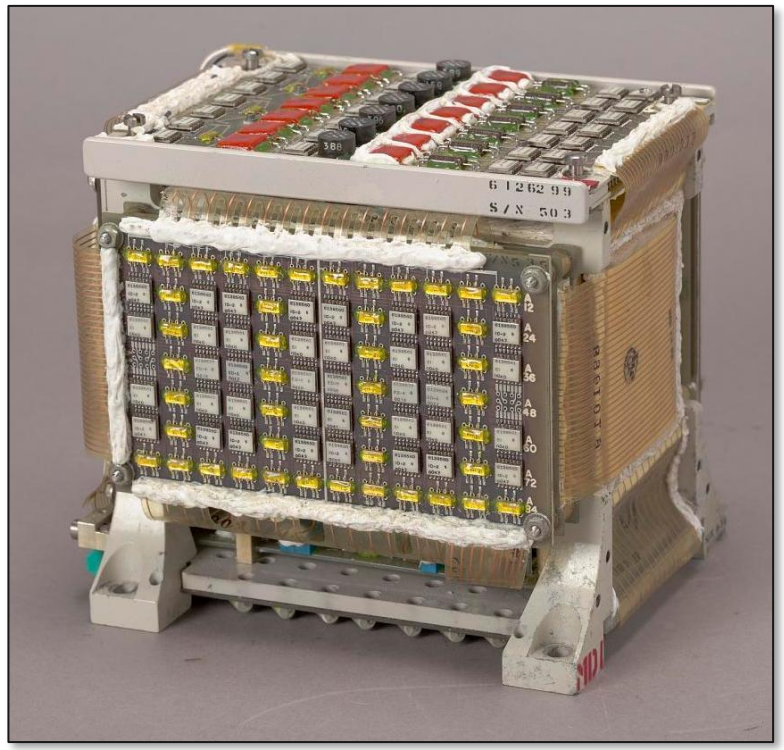
The IBM 2361 Core Storage Module housed 1MB or 2MB bytes of core memory, depending on the model. You can see the core planes (and the individual cores, i.e. bits) through the glass doors.



By Bubba73 - Own work, CC BY-SA 4.0, https://commons.wikimedia.org/w/index.php?curid=198495761

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www.Columbia.edu/cu/computinghistory/core.html

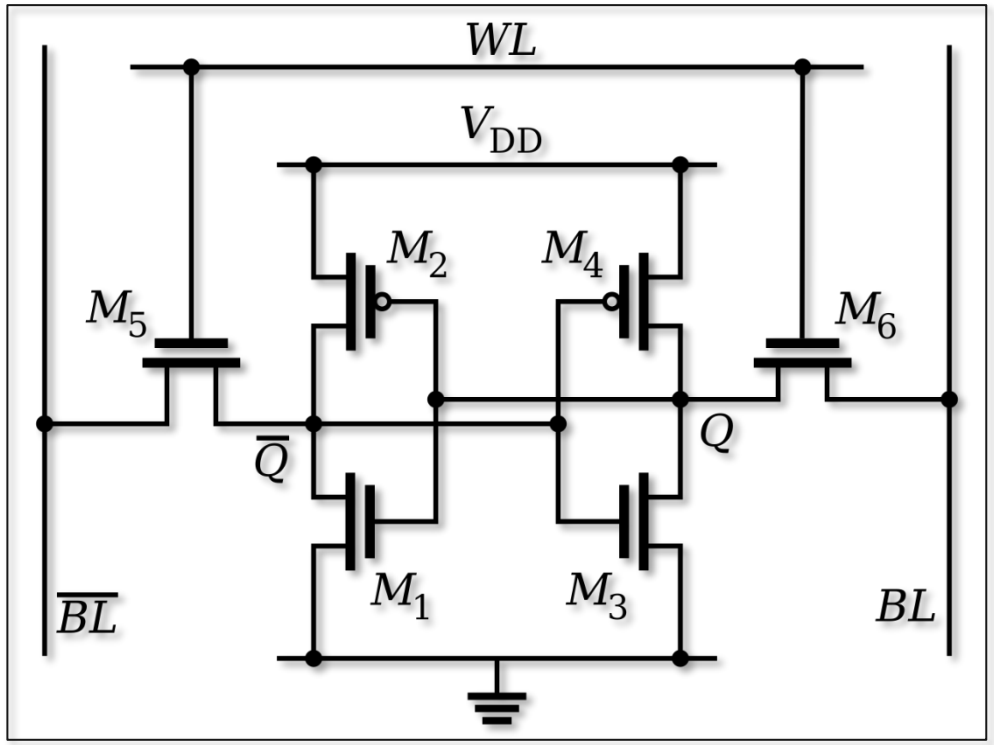
Launch Vehicle Digital Computer, Saturn V



Fair use, National Air and Space Museum
www.airandspace.si.edu/collection-media/NASM-AD8637935E702_001

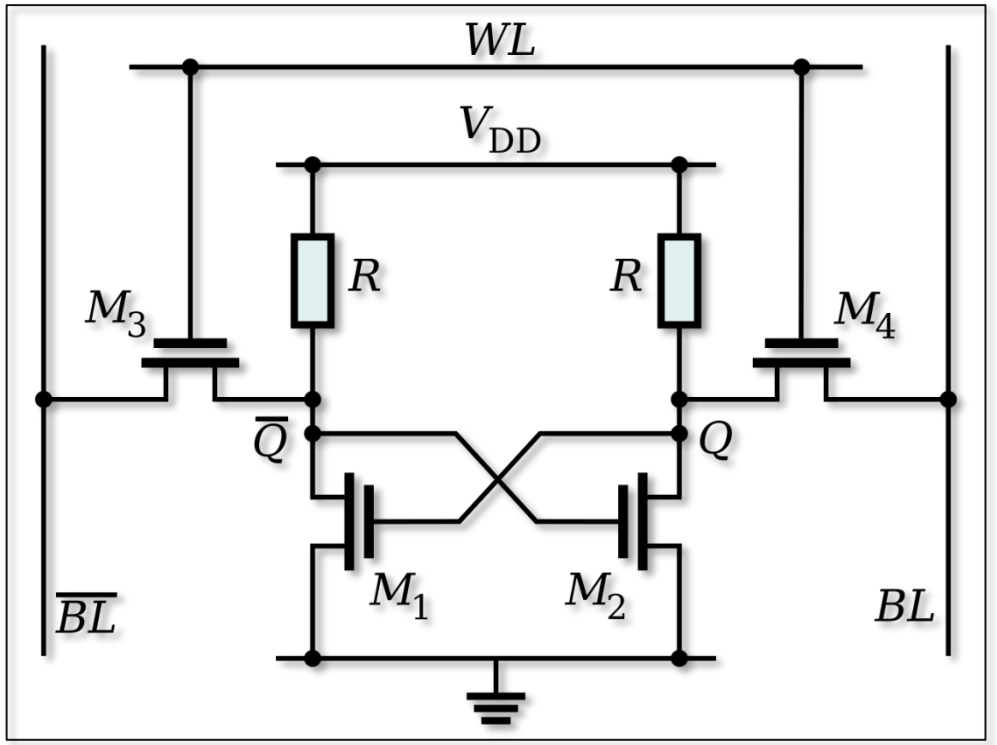
EE426 / 626 Microprocessor Applications

Static Random-Access Memory (SRAM), 6T



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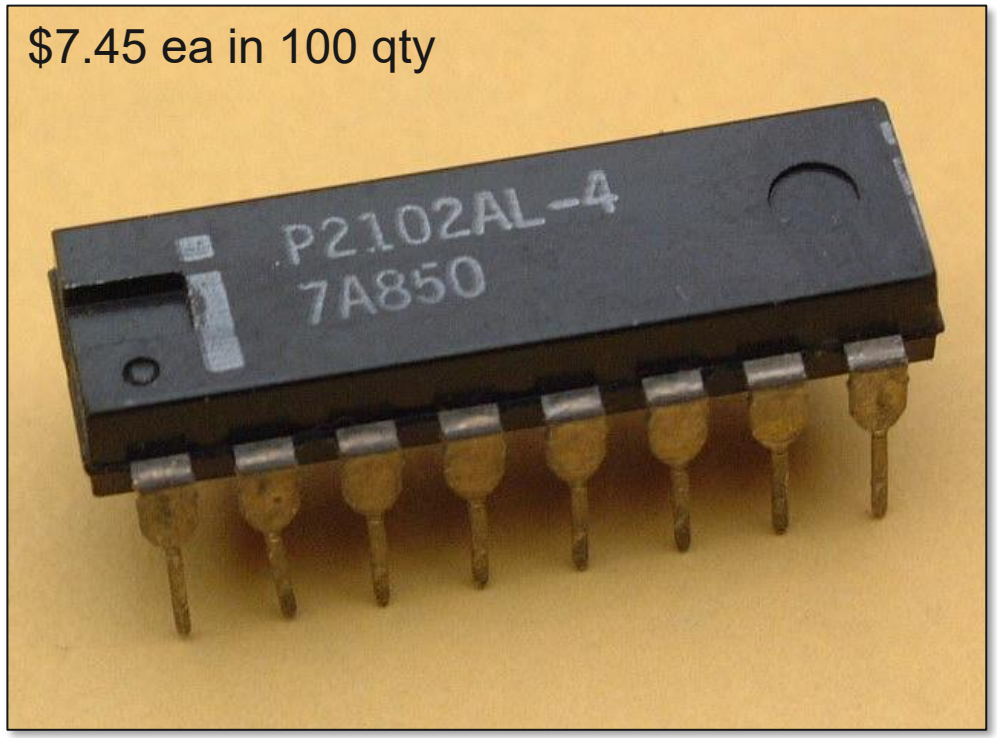
Static Random-Access Memory (SRAM), 4T



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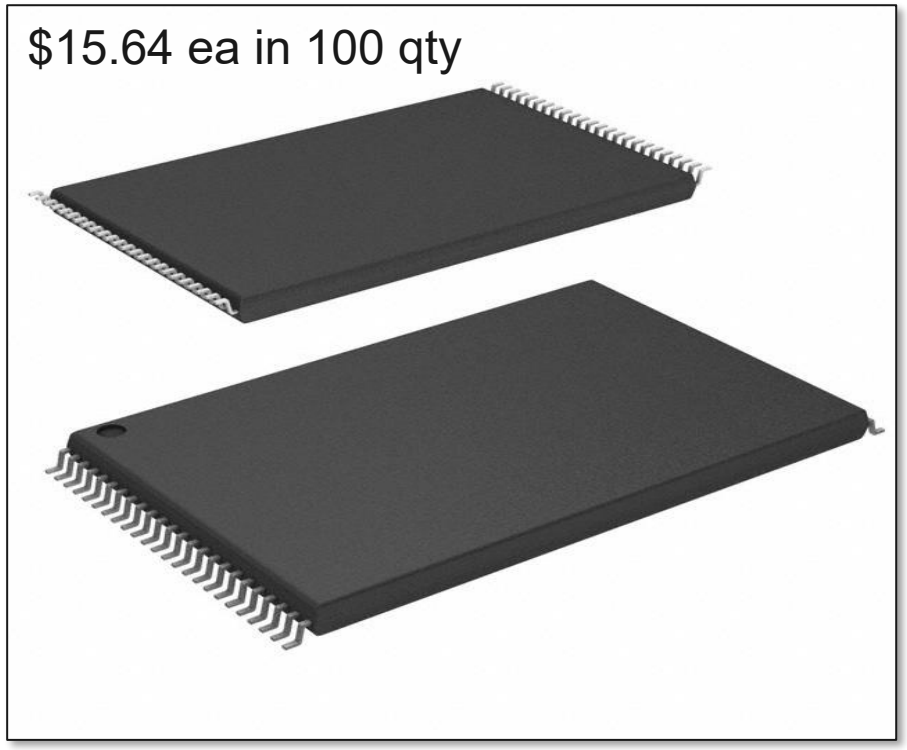
EE426 / 626 Microprocessor Applications

Intel 2102 (1024-bit SRAM)



Fair use, CPU-Zone, https://cpu-zone.com/2102/DSCF2015.JPG

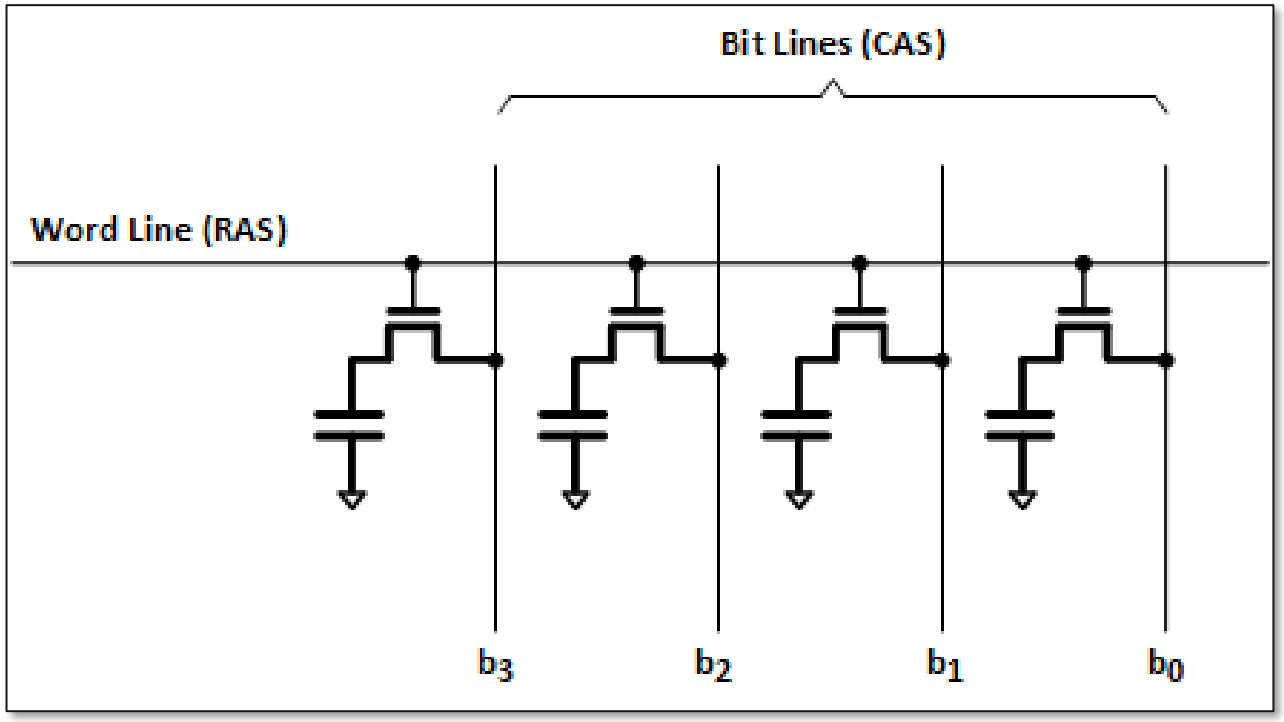
Cypress CY62167EV30 (2 MB x 8 SRAM)



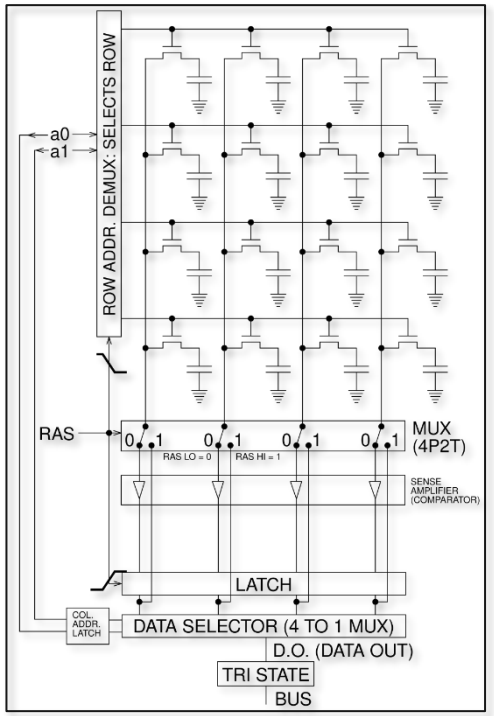
Fair use, https://digikey.com

EE426 / 626 Microprocessor Applications

Dynamic Random-Access Memory (DRAM) 1T1C



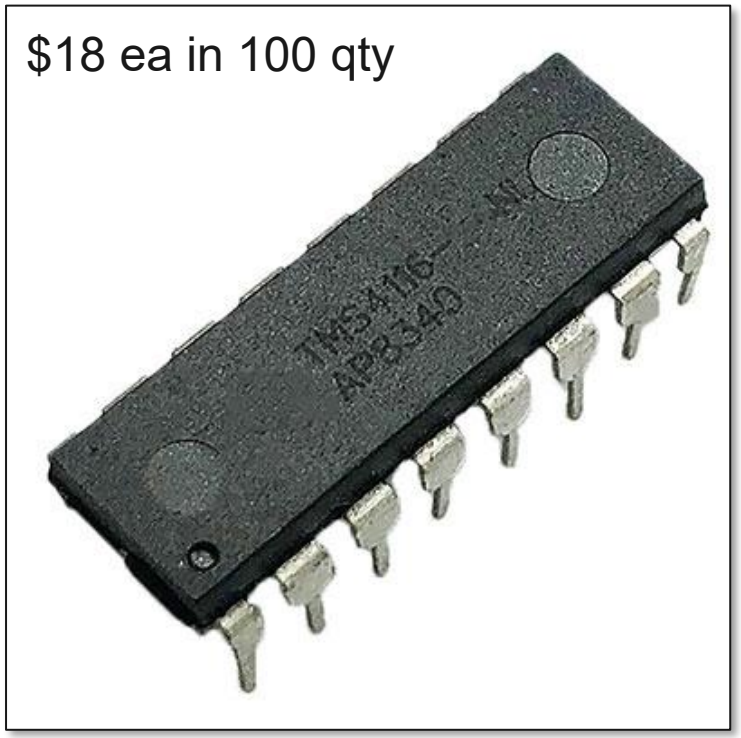
DRAM Array



By Encheart at English Wikipedia (Derivative SVG), Glogger at English Wikipedia (Original Image) - File:Square_array_of_mosfet_cells_read.png, CC BY-SA 3.0, https://commons.wikimedia.org/w/index.php?curid=63506811

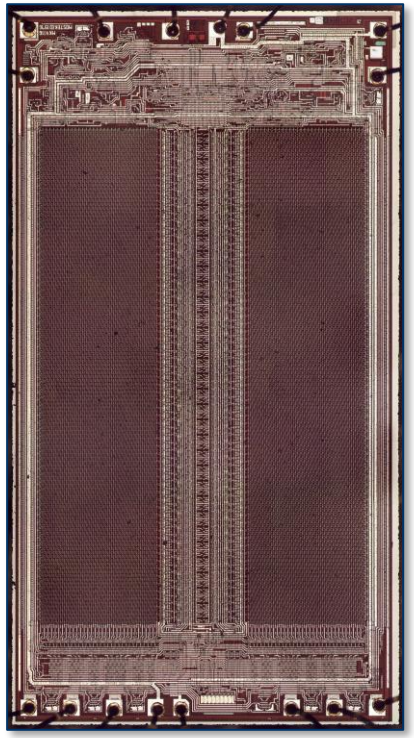
EE426 / 626 Microprocessor Applications

Mostek MK4116 (16k-bit DRAM)

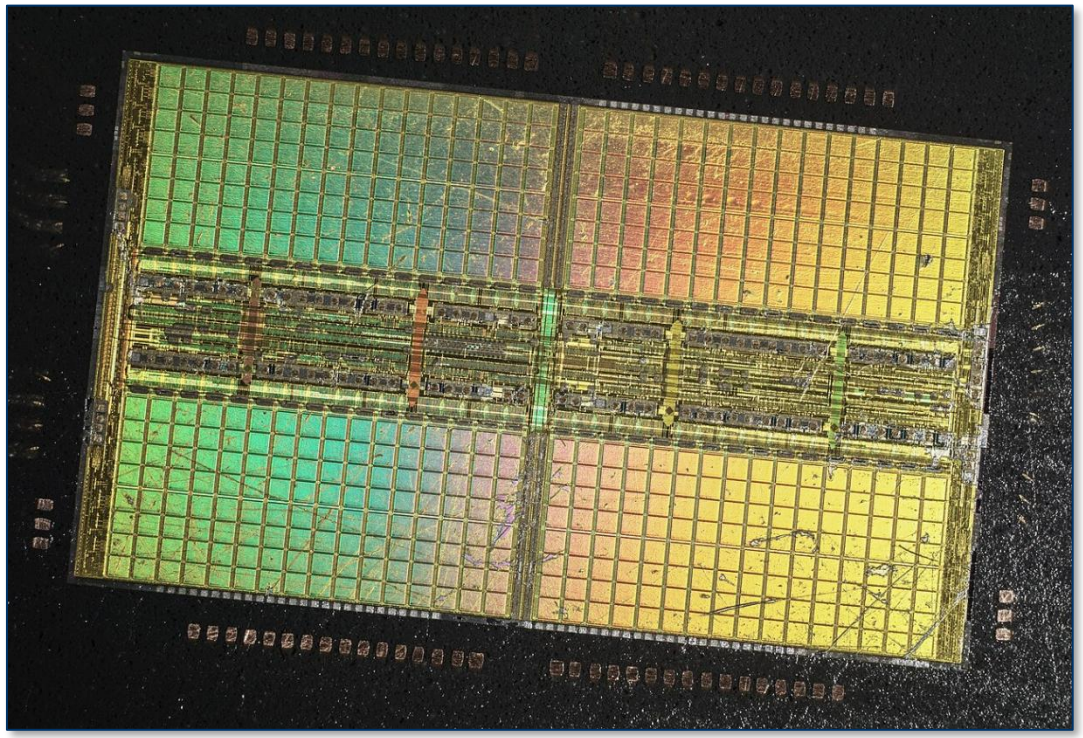


Photomicrograph by Ken Shirriff
https://rightto.com/2020/11/reverse-engineering-classic-mk4116-16.html

MK4116 Die



Samsung 64Mb DDR DRAM



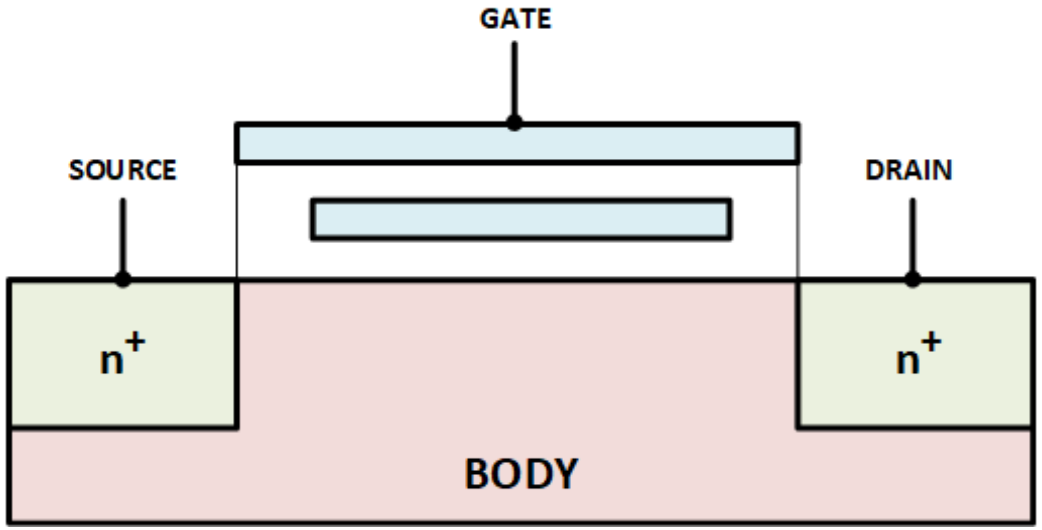
By Fritzchens Fritz
https://www.flickr.com/photos/130561288@N04/26941593496/,
CC0, https://commons.wikimedia.org/w/index.php?curid=95749782

EE426 / 626 Microprocessor Applications

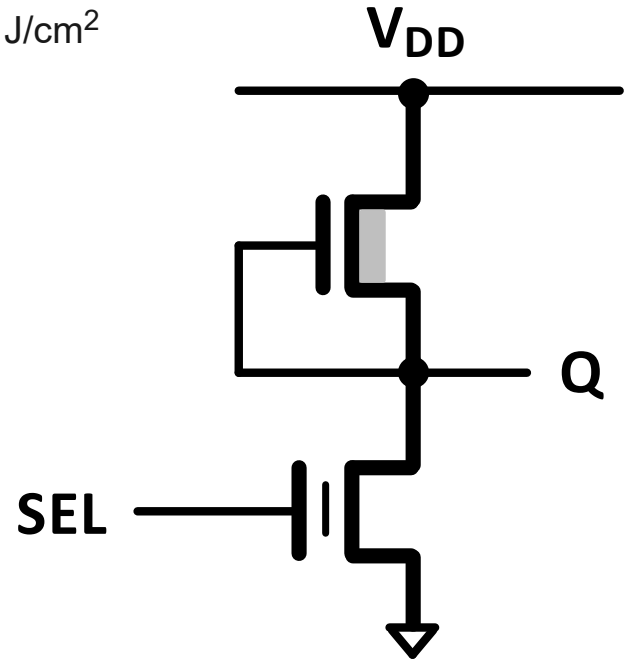
ROM / PROM / EPROM

Invented by Dov Frohman of Intel ~ 1972

EPROM – UV Erasable via Photoelectric effect, 250 nm < λ < 400 nm, 15 J/cm^2



Floating Gate Transistor – Proposed at Bell Labs 1967

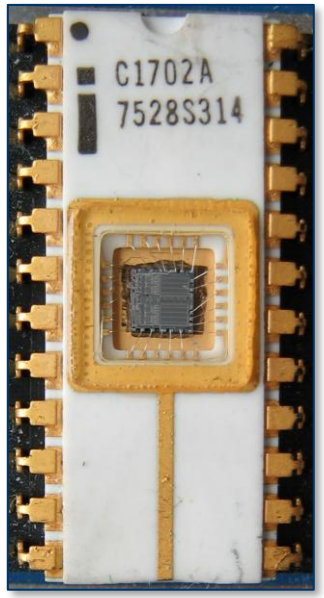


Conceptual Diagram of a 1-bit Cell

EE426 / 626 Microprocessor Applications

ROM / PROM / EPROM

Intel 1702 (256 × 8-bit)



By Author : Poil 01:10, 17 Apr 2005 (UTC) - Author personnal collection., CC BY-SA 3.0, https://commons.wikimedia.org/w/index.php?curid=97658

ST Microelectronics
M27C256B-12 (32k × 8-bit)



By Bill BertramPixel8Photo by:Bill BertramCamera:Fujifilm FinePix F106.3 Megapixels 3x Optical zoom - Own work, CC BY-SA 2.5, https://commons.wikimedia.org/w/index.php?curid=156358217

Atmel
AT27C010-70 (128k × 8-bit)



By © Raimond Spekking / CC BY-SA 4.0 (via Wikimedia Commons), CC BY-SA 4.0, https://commons.wikimedia.org/w/index.php?curid=50745759

EE426 / 626 Microprocessor Applications

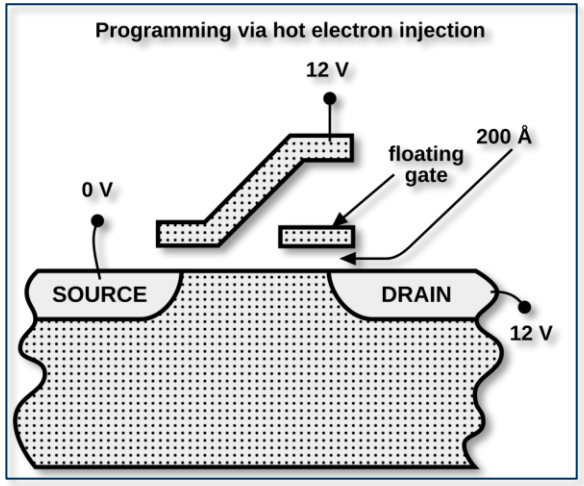
EEPROM or E^{2}PROM – Electrically Erasable Programmable Read-Only Memory

1976 to 1978 – George Perlegos at Intel (Develops 2816 2k × 8 EEPROM)

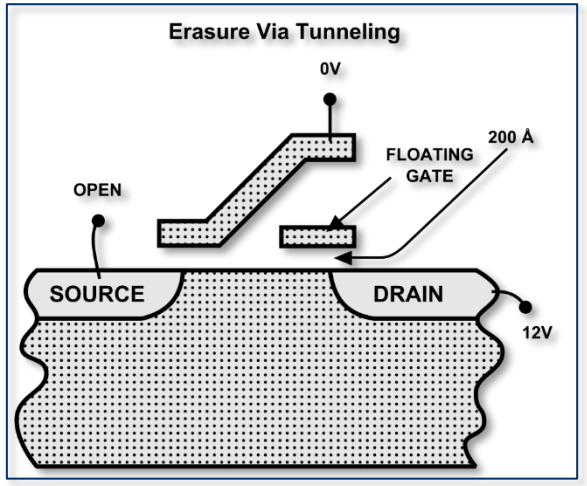
1980 – Floating Gate Tunnel Oxide (FLOTOX)

1981 – Perlegos leaves Intel to found Seeq Technology

1984 – Perlegos leaves Seeq Technology to found Atmel (and acquires Seeq)



By Original: Harald wehner Vector: David W. - Own work based on: Flash-Programming.png, Public Domain, https://commons.wikimedia.org/w/index.php?curid=1363113

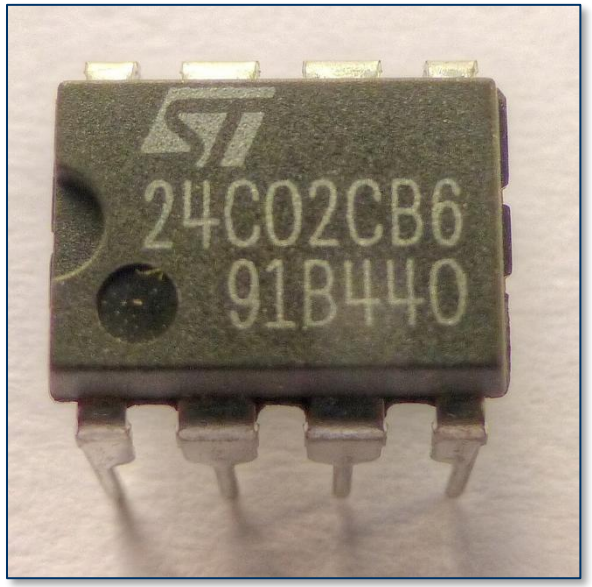


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EE426 / 626 Microprocessor Applications

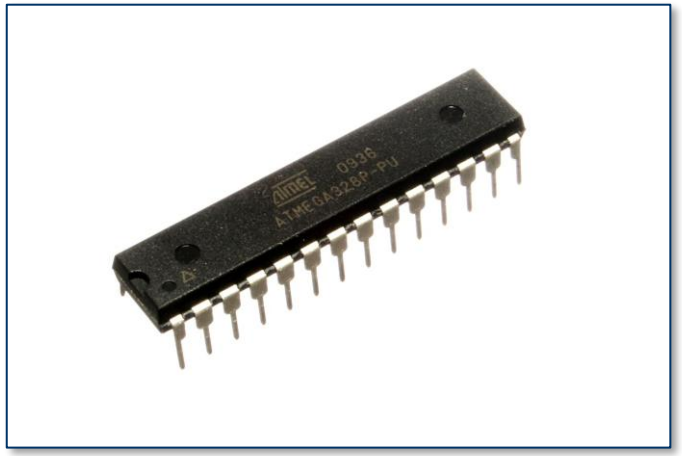
EEPROM or E^2PROM

ST Microelectronics
M24C02 (256 × 8-bit)



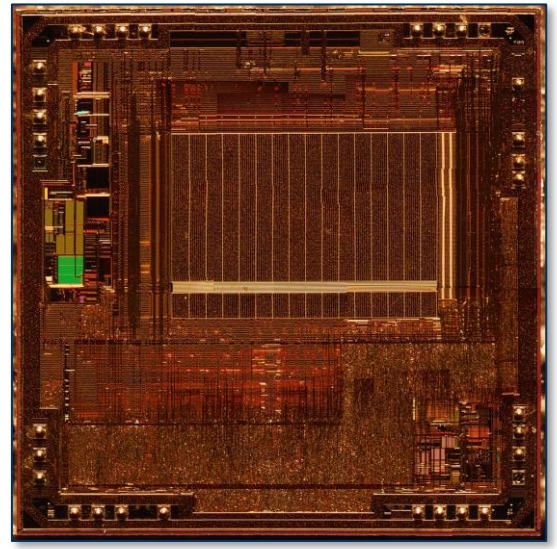
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Atmel (Microchip Technology)
ATMega328 MCU



By oomlout - Flickr: ATMEGA328 - IC-ATM328-01, CC BY-SA 2.0,
https://commons.wikimedia.org/w/index.php?curid=23315198

Atmel (Microchip Technology)
ATMega328 Die



By BregesT65421354 - Own work, CC BY-SA 4.0,
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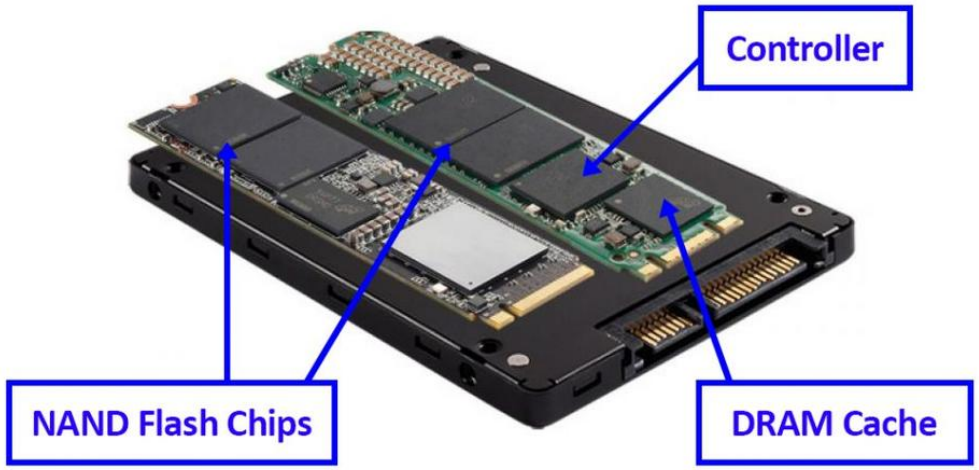
EE426 / 626 Microprocessor Applications

FLASH (A type of EEPROM)

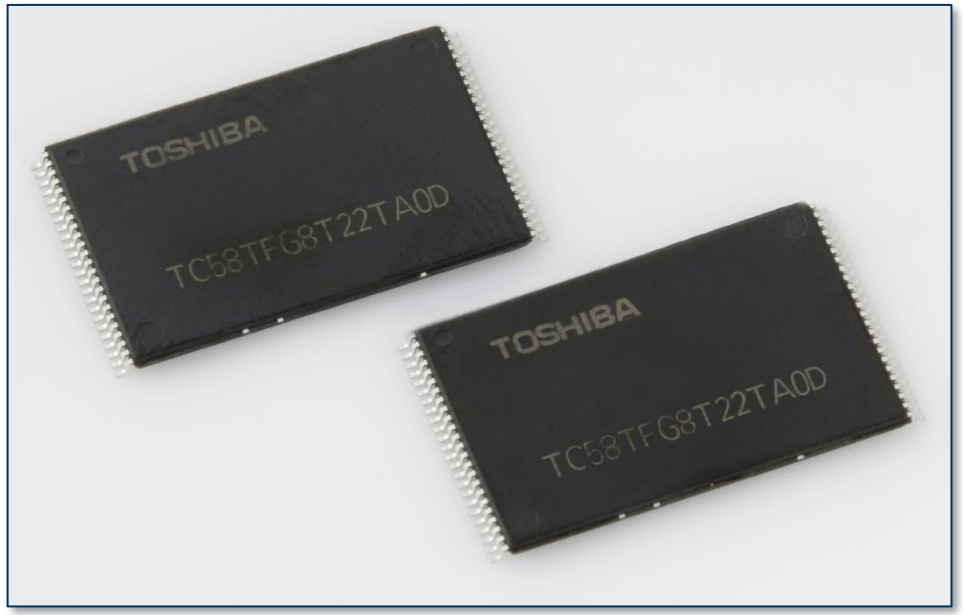
Invented by Fujio Masuoka at Toshiba ~ 1980

NOR flash – Slower, less dense, but Random Access

NAND Flash – Faster, cheaper, dense, but Block Access

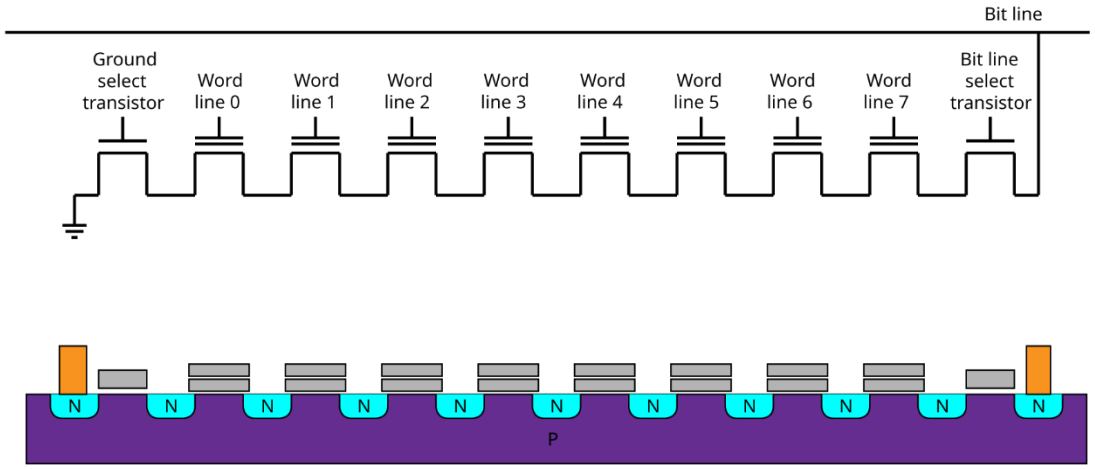


Toshiba / SanDisk 256Gb 48-layer NAND flash

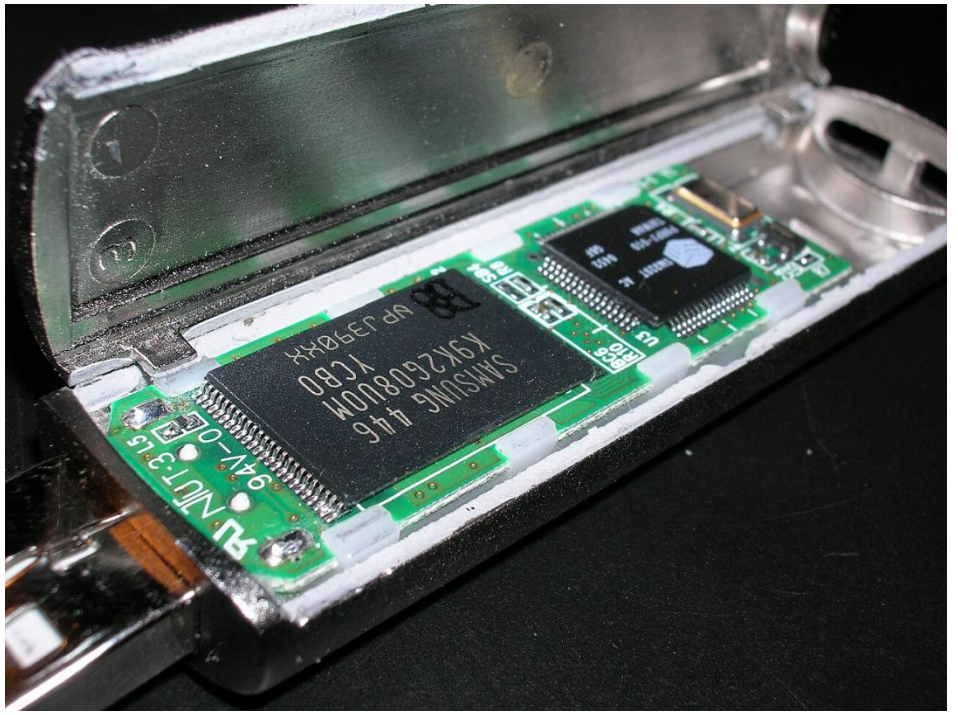


EE426 / 626 Microprocessor Applications

NAND FLASH – Most common for economical bulk storage



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By Nrbelex, CC BY-SA 3.0, https://commons.wikimedia.org/w/index.php?curid=179956

EE426 / 626 Microprocessor Applications

STM32 Memory

Internal Memory

- FLASH — main program NOR Flash, up to 512KB @ 0x0800_0000, erasable by sector
- SYSMEM — System memory / boot ROM (factory bootloader), @ 0x1FFF_0000
- OTP — One-Time Programmable area, 528 bytes @ 0x1FFF_7800
- Option Bytes — NOR FLASH Configuration @ 0x1FFF_C000, never a linker target
- SRAM1 — main system RAM, 112KB @ 0x2000_0000
- SRAM2 — additional RAM, 16KB, contiguous with SRAM1 (128KB total)
- CCMRAM — Core-Coupled Memory (CCM), 64KB @ 0x1000_0000
- BKPSRAM — Backup SRAM, 4KB @ 0x4002_4000, VBAT-backed domain

External Memory, via Flexible Memory Controller (FMC) peripheral

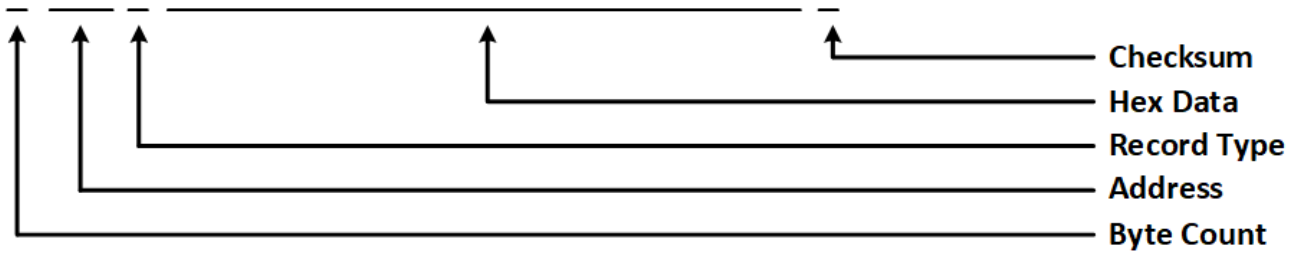
- NOR — NOR Flash, Bank 1 (0x6000_0000, 4 sub-banks)
- PSRAM — Pseudo SRAM (PSRAM), Bank 1
- EXTSRAM — Asynchronous SRAM, Bank 1
- NAND — NAND Flash, Bank 3 (0x8000_0000), hardware ECC up to 8KB
- SDRAM — Synchronous DRAM (SDRAM), SDRAM Bank 1/Bank 2 (0xC000_0000 / 0xD000_0000)



EE426 / 626 Microprocessor Applications

Intel HEX Format

```
:02000004040800F2
:10000000A00020550D0008A50D0008A50D000852
:10001000A50D0008A50D0008A50D000800000000B2
:10002000000000000000000000000000A50D000816
:10003000A50D00080000000A50D00083D4D0008BA
:10004000A50D0008A50D0008A50D00082141000818
:10005000A50D0008A50D0008A50D0008A50D0008B8
```



Record Types (for EE426/626)

00 Data

```
:04001000FFFFFFFFFE
```

01 End of File

```
:00000001FF
```

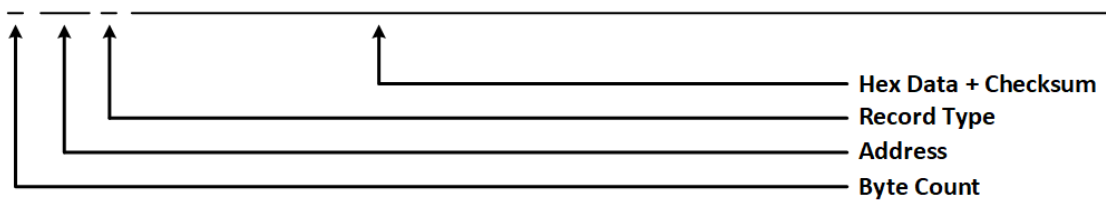
04 Extended Linear Address

```
:02000004040400F6
```

EE426 / 626 Microprocessor Applications

Intel HEX Format – Example

```
:19010000556E697665727369747920 6F66204E6576616461205 2656E6F8C
:2302000045453432362F363236204D6963726F70726F6365737320417070 6C69636174696F6E73C8
:09030000476F20506163636B21215C
:00000001FF
```




```
0100 55 6E 69 76 65 72 73 69 'Universi'
0108 74 79 20 6F 66 20 4E 65 'ty of Ne'
0110 76 61 64 61 20 52 65 6E 'vada Ren'
0118 6F FF FF FF FF FF FF FF 'o.......'

0200 45 45 34 32 36 2F 36 32 'EE426/62'
0208 36 20 4D 69 63 72 6F 70 '6 Microp'
0210 72 6F 63 65 73 73 20 41 'rocess A'
0218 70 70 6C 69 63 61 74 69 'pplicati'
0220 6F 6E 73 FF FF FF FF FF 'ons.....'

0400 47 6F 20 50 61 63 6B 21 'Go Pack!'
0408 21 FF FF FF FF FF FF FF '!.......'
```

EE426 / 626 Microprocessor Applications

Intel HEX Format – Example

M27C512 EPROM



Dataman 48Pro+ Programmer

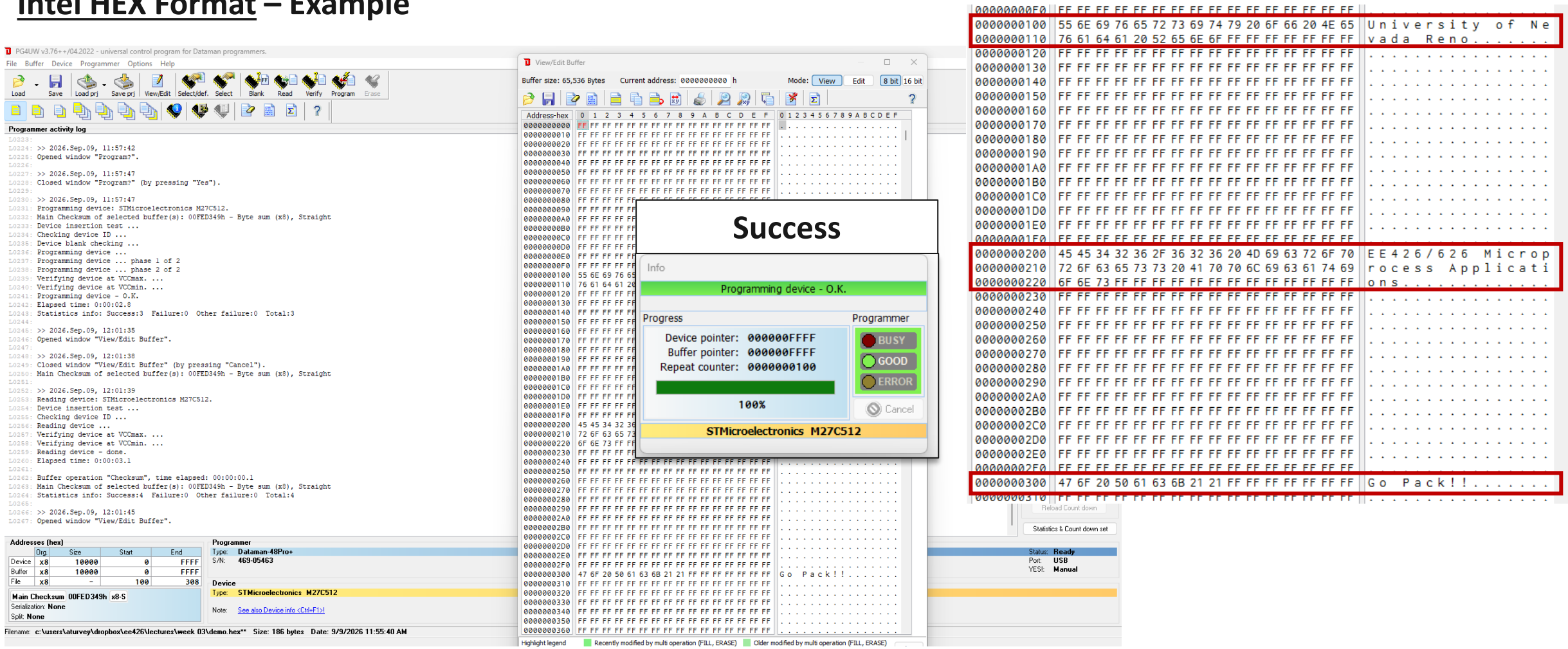


PE-140T UV Lamp



EE426 / 626 Microprocessor Applications

Intel HEX Format – Example



EE426 / 626 Microprocessor Applications

C Programming / Memory Sections / Segue to the Linker Script

```
int    g_counter = 42;                    // .data    -> SRAM1    (copied from FLASH at startup)
int    g_flag;                            // .bss     -> SRAM1    (zero-init at startup)
const  uint8_t LUT[4] = {1,2,3,4};        // .rodata  -> FLASH    (read-only, stays in FLASH)
static int s_state;                       // .bss     -> SRAM1    (static == global-uninit rules)

void foo(void) {
    int x;                                //  stack   -> SRAM1    (no linker symbol, lives inside
}                                         //                       the linker script's stack region)

uint8_t framebuf[320*240]
    __attribute__((section(".sdram")));   // .sdram   -> External SDRAM (FMC Bank 1 — needs a
                                          //                      matching MEMORY region +
                                          //                      SECTIONS entry, compiler won't
                                          //                      place it there by default)
```

